



Joint
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two discrete
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Correlation
coefficient

Numeric
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of the sample

Visualisation of
the sample

Kolmogorov's
strong law of
large numbers,
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Lecture notes for Applies statistics, Probability theory

Lecture 5.

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Presentation Overview

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Joint distribution of two discrete random variables



Definition

Let X and Y be discrete random variables. Let the range of X be $x_1, x_2, \dots$, the range of Y be $y_1, y_2, \dots$. Then the joint distribution of X and Y is

$$p_{ij} = P(X = x_i, Y = y_j), \quad i, j = 1, 2, \dots$$

We see that these numbers are non-negative and their sum is equal to 1, that is

$$\sum_{i=1}^{\infty} \sum_{j=1}^{\infty} p_{ij} = 1.$$



Marginal distributions

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Definition

The distribution of X is

$$p_{i\cdot} = P(X = x_i) = \sum_{j=1}^{\infty} p_{ij}$$

and the distribution of Y is

$$p_{\cdot j} = P(Y = y_j) = \sum_{i=1}^{\infty} p_{ij}.$$

These are called the two marginal distributions. These numbers are non-negative and

$$\sum_{i=1}^{\infty} p_{i\cdot} = 1, \quad \sum_{j=1}^{\infty} p_{\cdot j} = 1.$$



Contingency table

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$\begin{matrix} Y \\ X \end{matrix}$	y_1	y_2	y_3	$\dots$	Σ
x_1	p_{11}	p_{12}	p_{13}	$\dots$	$p_{1.}$
x_2	p_{21}	p_{22}	p_{23}	$\dots$	$p_{2.}$
x_3	p_{31}	p_{32}	p_{33}	$\dots$	$p_{3.}$
$\dots$	$\dots$	$\dots$	$\dots$	$\dots$	$\dots$
Σ	$p_{.1}$	$p_{.2}$	$p_{.3}$	$\dots$	1

Fig. 1: Contingency table



Independence of DRV

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Definition

X and Y are called independent if

$$P(X = x_i, Y = y_j) = P(X = x_i)P(Y = y_j), i, j = 1, 2, \dots$$

That is

$$p_{ij} = p_{i \cdot} \cdot p_{\cdot j}, \quad \forall i, j.$$



Joint distribution function of abs. cont. random variables

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Definition

Let X and Y be random variables. Then the function

$$F(x, y) = P(X < x, Y < y), \quad x, y \in \mathbb{R},$$

is called the joint cumulative distribution function (joint CDF) of X and Y .

Let the joint CDF of X and Y be $F(x, y)$. Then the CDF of X is $F_X(x) = \lim_{y \rightarrow \infty} F(x, y)$ and the CDF of Y is $F_Y(y) = \lim_{x \rightarrow \infty} F(x, y)$. F_X and F_Y are called the marginal CDF's of $F(x, y)$.



Definition

The joint distribution of X and Y is called absolutely continuous if there exists a two variable function f so that the joint CDF F is

$$F(x, y) = \int_{-\infty}^x \int_{-\infty}^y f(u, v) dv du, \quad x, y \in \mathbb{R}$$

for all $x, y \in \mathbb{R}$. f is called the joint probability density function (joint PDF) of X and Y .

Let the joint PDF of X and Y be f . Then both X and Y have PDF and they are

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy, x \in \mathbb{R}; \quad f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx, y \in \mathbb{R}.$$

f_X and f_Y are called the marginal PDF's.



Definition

X and Y are called independent if their joint CDF is the product of the two marginal CDF's, that is

$$F_{X,Y}(x,y) = F_X(x) \cdot F_Y(y), \quad x,y \in \mathbb{R}.$$

Let the joint distribution of X and Y be absolutely continuous. Then X and Y are independent if and only if their joint PDF is the product of the two marginal PDF's, that is

$$f_{X,Y}(x,y) = f_X(x) \cdot f_Y(y), \quad x,y \in \mathbb{R}.$$



Definition

Let X and Y be random variables, $\text{Var}X < \infty$, $\text{Var}Y < \infty$. The covariance of X and Y is

$$\text{cov}(X, Y) = \mathbb{E}((X - \mathbb{E}X)(Y - \mathbb{E}Y)).$$

To calculate the covariance use

$$\text{cov}(X, Y) = \mathbb{E}(XY) - \mathbb{E}X \cdot \mathbb{E}Y.$$



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Correlation coefficient



Correlation coefficient

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The strength on the connection on two random variables can be defined by the correlation coefficient.

Definition

Let $\mathbb{D}^2 X > 0$ and $\mathbb{D}^2 Y < \infty$. the correlation coefficient of X and Y is:

$$\text{corr}(X, Y) = \frac{\text{cov}(X, Y)}{\mathbb{D}X \mathbb{D}Y}.$$

If $\text{corr}(X, Y) = 0$, then X and Y are not correlated.

To calculate the correlation use

$$\text{corr}(X, Y) = \frac{\mathbb{E}(XY) - \mathbb{E}X \cdot \mathbb{E}Y}{\mathbb{D}X \mathbb{D}Y}.$$



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Important note

If two random variables are independent, then they are not correlated, but if two random variables are not correlated does not mean, that they are independent

Example(source: Wikipedia)

If X and Y are two random variables, such that $P(X = 0) = 1/2$, $P(X = 1) = 1/2$ and Y is independent from X and, $P(Y = -1) = 1/2$, $P(Y = 1) = 1/2$ and let $U = XY$.

$$\mathbb{E}U = \mathbb{E}XY = \mathbb{E}X \cdot \mathbb{E}Y = \mathbb{E}X \cdot 0 = 0$$

$$\begin{aligned} \text{cov}(U, X) &= \mathbb{E}((U - \mathbb{E}U)(X - \mathbb{E}X)) = \mathbb{E}(U(X - 1/2)) = \\ &= \mathbb{E}(X^2Y - (1/2)XY) = \mathbb{E}((X^2 - (1/2)X)Y) = \mathbb{E}((X^2 - (1/2)X))\mathbb{E}Y = 0 \end{aligned}$$

Therefore, U and X are not correlated.



Example

If X and U are independent, then $P(U = a|X = b) = P(U = a), \forall a, b \in \mathbb{R}$.

$$P(U = 1|X = 0) = P(XY = 1|X = 0) = 0$$

$$P(U = 1) = P(XY = 1) = 1/4$$

Therefore, U and X are not independent.



Correlation coefficient

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Theorem

(1) For any random variables ξ and η : $-1 \leq \text{corr}(\xi, \eta) \leq 1$.

(2) $|\text{corr}(\xi, \eta)| = 1$ if and only if

$$\eta = a\xi + b$$

for some $a \neq 0$, $a, b \in \mathbb{R}$.



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Definition

The arithmetic mean of a set of observed data is equal to the sum of the numerical values of each observation, divided by the total number of observations.

$$\bar{x} = \frac{1}{n} \left(\sum_{i=1}^n x_i \right) = \frac{x_1 + x_2 + \cdots + x_n}{n}$$



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Definition

m is called the median of ξ if:

$$P(\xi < m) \leq 1/2, \quad P(\xi > m) \leq 1/2$$



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Definition

The mode of a sample is the element that occurs most often in the sample.



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Histogram

Definition

The histogram consists of n rectangles whose area is proportional to the frequency of an element in the sample and their width is equal to the class interval.

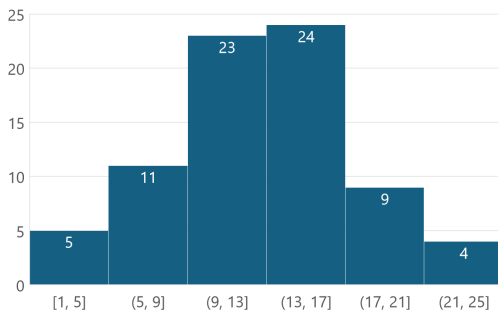


Fig. 2: Histogram



Box plot

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Definition

Box plot is a method for demonstrating graphically the locality, spread and skewness groups of numerical data through their quartiles.

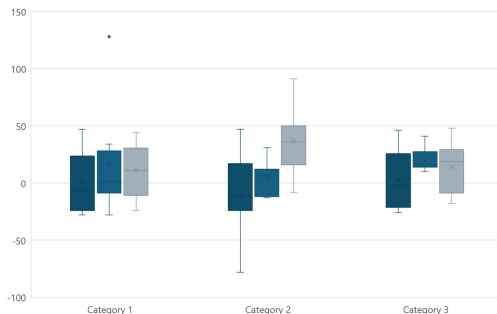


Fig. 3: Box plot



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Theorem

Let $X_1, X_2, \dots$ be independent and identically distributed random variables with $\mathbb{E}|X_i| < \infty$. Let $m = \mathbb{E}X_i$ be their expectation. Then

$$\lim_{n \rightarrow \infty} \frac{S_n}{n} = m \quad \text{almost surely.}$$

The meaning of the above limit is the following

$$P\left(\lim_{n \rightarrow \infty} \frac{S_n}{n} = m\right) = 1$$



Visualization of SLLN

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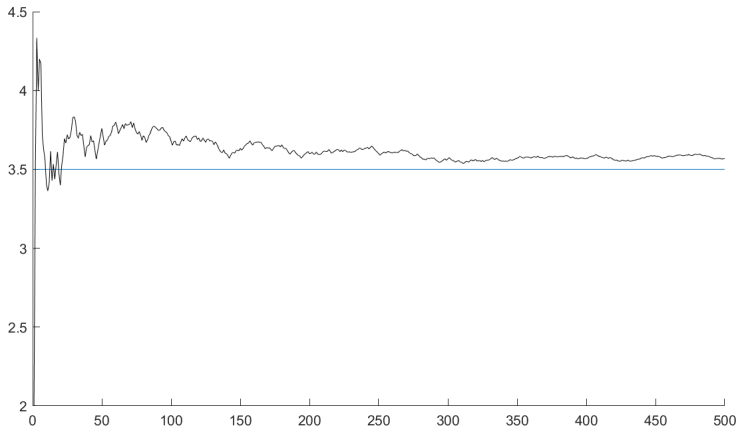


Fig. 4: Visualization of SLLN, rolling a dice 500 times

blue line - theoretical expected value, black line - empirical result